PVSYST
PHOTOVOLTAIC
SOFTWARE

PVsyst TRIAL

PVsyst - Simulation report

Grid-Connected System

Project: Manaus

Variant: New simulation variant
No 3D scene defined, no shadings
System power: 5.40 kWp
Manaus - Brazil

Author

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PVsyst V8.1.7
VC0, Simulation date:
16/10/26 21:55
with V8.1.7

Project summary

Geographical Site
Manaus
Brazil

Situation

Latitude	-3.1316 °(S)
Longitude	-59.9825 °(W)
Altitude	43 m
Time zone	UTC-4

Project settings
Albedo 0.20

Weather data
Manaus
NASA POWER satellite data 2001-2020 - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1
Horizontal plane

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	9 units	Nb. of units	1 unit
Pnom total	5.40 kWp	Total power	5 kWac
		Pnom ratio	1.08

Results summary

Simulation step	Hourly				
Produced Energy	7834.6 kWh/year	Specific production	1451 kWh/kWp/year	Perf. Ratio PR	85.09 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Simulation step
Hourly

Orientation #1
Horizontal plane

Models used

Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	JKM-600N-72HL4-V
(Original PVsyst database)	
Unit Nom. Power	600 Wp
Number of PV modules	9 units
Nominal (STC)	5.40 kWp
Modules	1 unit x 9 In series
At operating cond. (50°C)	
Pmpp	5.01 kWp
U mpp	367 V
I mpp	14 A

Inverter

Manufacturer	Generic
Model	MIN 5000TL-X
(Original PVsyst database)	
Unit Nom. Power	5.00 kWac
Number of inverters	1 unit
Total power	5.0 kWac
Operating voltage	80-550 V
Pnom ratio (DC:AC)	1.08
Power sharing within this inverter	

Total PV power

Nominal (STC)	5 kWp
Total	9 modules
Module area	23.2 m²
Cell area	21.5 m²

Total inverter power

Total power	5 kWac
Number of inverters	1 unit
Pnom ratio	1.08

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 440 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

Strings Mismatch loss
Loss Fraction 0.15 %

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

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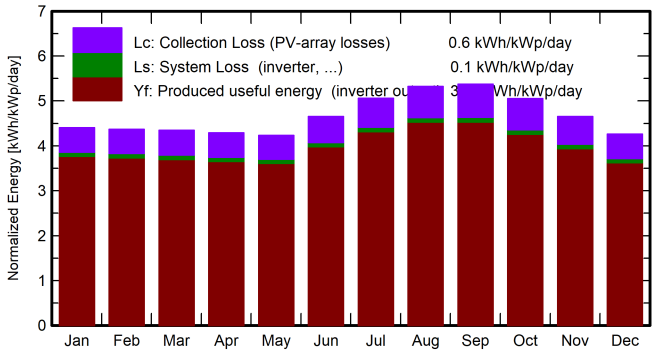
Main results

Simulation step
Hourly

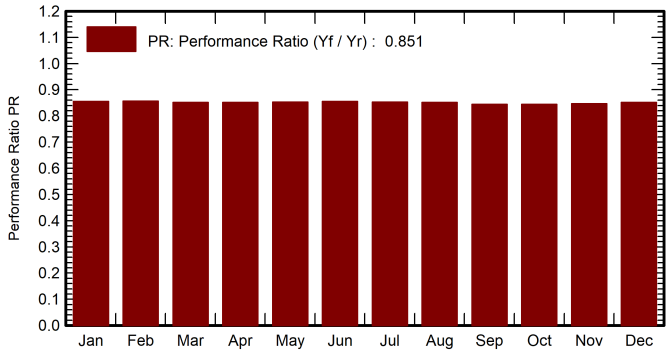
System Production

Produced Energy	7834.6 kWh/year	Specific production	1451 kWh/kWp/year
		Perf. Ratio PR	85.09 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	136.7	73.91	26.47	136.6	133.0	647.1	631.5	0.856
February	122.4	67.40	26.21	122.3	118.9	579.3	565.3	0.856
March	134.9	73.15	26.17	134.9	131.4	635.9	619.8	0.851
April	128.8	70.05	26.13	128.8	125.4	607.5	592.1	0.852
May	131.3	67.81	26.18	131.3	127.6	620.2	604.6	0.853
June	139.8	57.87	26.00	139.7	135.9	660.8	645.1	0.855
July	156.9	59.93	26.27	156.9	152.7	739.4	722.6	0.853
August	165.1	66.26	27.55	165.1	161.0	775.8	758.6	0.851
September	161.2	70.40	28.95	161.2	157.4	751.5	734.5	0.844
October	156.7	79.41	29.16	156.6	152.8	730.8	714.1	0.845
November	139.8	75.41	28.41	139.7	136.0	654.6	639.1	0.847
December	132.2	72.36	27.09	132.1	128.1	622.3	607.3	0.851
Year	1705.7	833.96	27.05	1705.1	1660.1	8025.4	7834.6	0.851

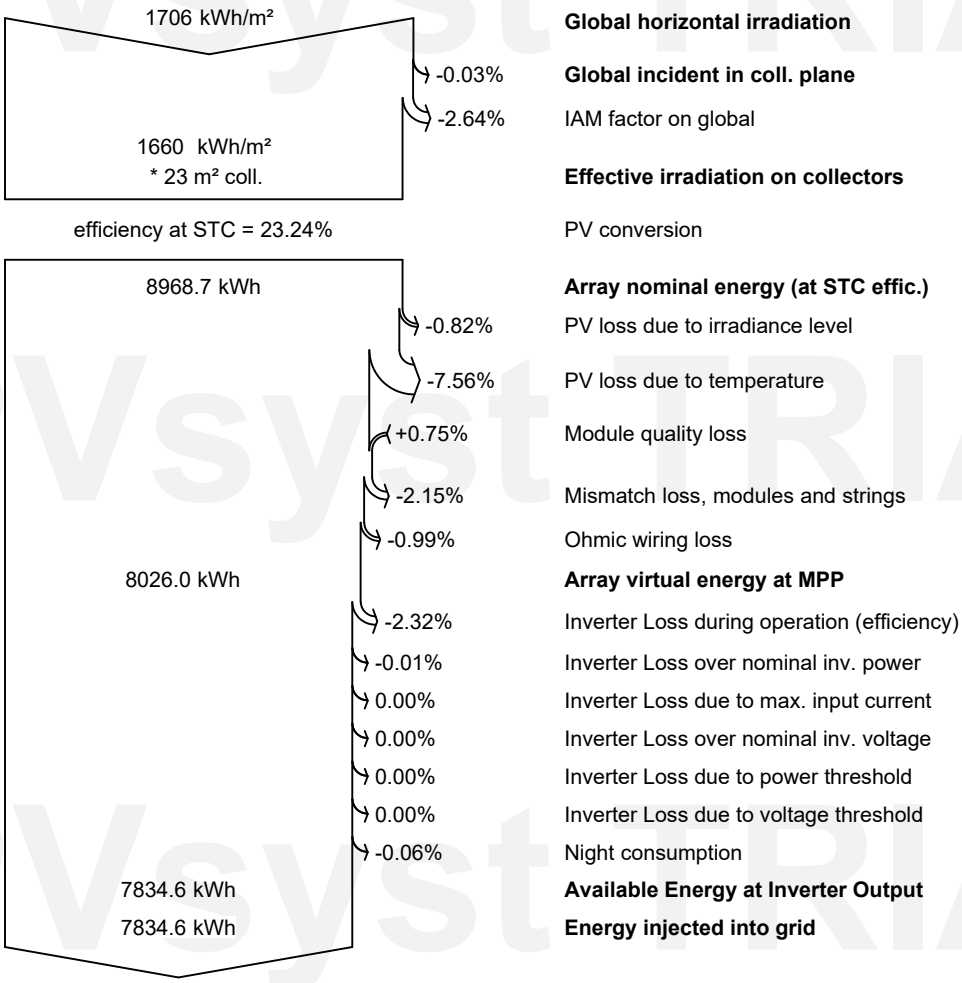
Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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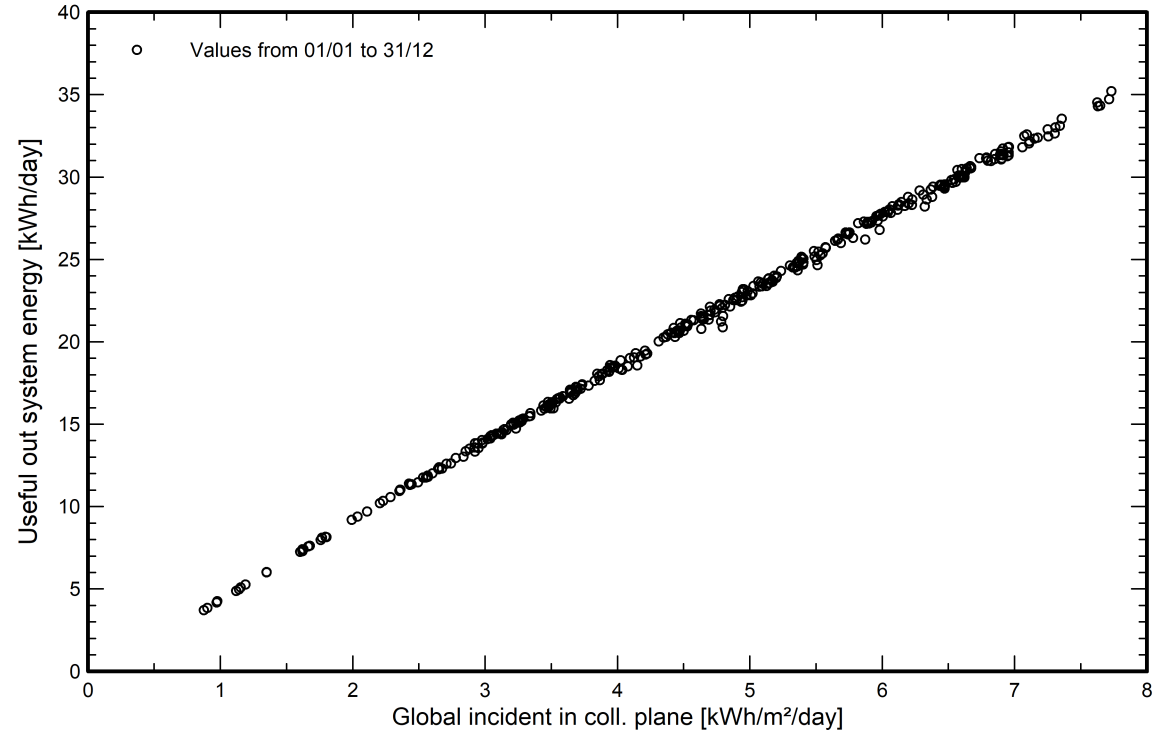
Loss diagram



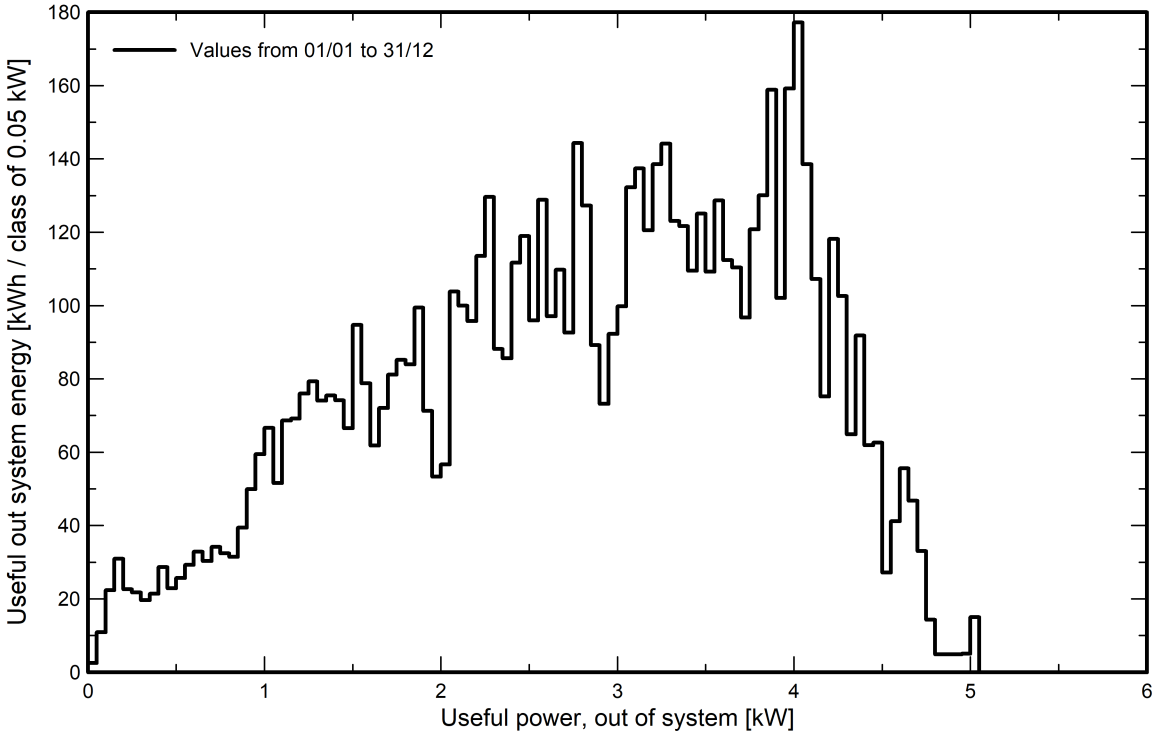
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Predef. graphs

Daily Input/Output diagram

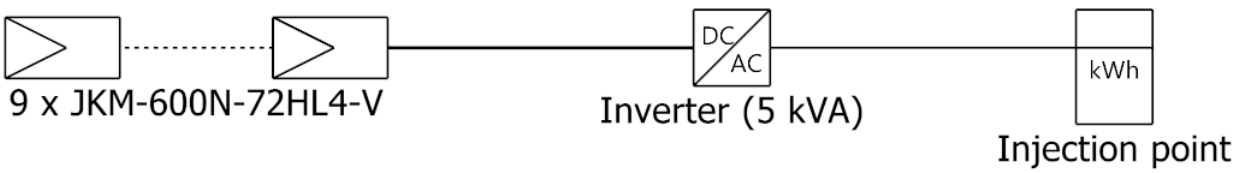


System Output Power Distribution



Single-line diagram

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PV module	JKM-600N-72HL4-V
Inverter	MIN 5000TL-X
String	9 x JKM-600N-72HL4-V

	Manaus	
	VC0 : New simulation variant	16/10/26